Residual certificates for the Erdős–Straus conjecture: exact solvability criteria, a sieve bound, and computations to 10^{10}

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Abstract

The Erdős–Straus conjecture asserts that $4/n = 1/a + 1/b + 1/c$ is solvable in positive integers for every $n \geq 2$. The open cases concentrate on primes in the six residue classes $\{1, 121, 169, 289, 361, 529\}$ mod 840 (*hard primes*). We study the *residual method*: for $R \equiv 3 \pmod 4$ set $a = (p + R)/4$, $m = pa$; a solution with first denominator a exists if and only if some divisor $k \mid m^2$ satisfies $k \equiv -m \pmod R$. Our main theoretical results are: (i) *exact* solvability criteria for the residuals $R = 3$, 7, and 11 at hard primes, the latter two established by machine-verified finite case analyses; (ii) a meta-theorem showing such a finite exact criterion exists and is computable for every fixed R; (iii) an unconditional chain of sieve bounds, via a machine-verified *support-bound lemma* that makes every failure at each prime residual a sifting condition of dimension $\geq \frac{1}{2}$: the number of hard primes $p \leq x$ solved by no residual in $\{3, 7\}$ is $O\big(x/(\log x)^2\big)$, by no residual in $\{3, 7, 11\}$ is $O\big(x/(\log x)^{5/2}\big)$, and — with the lemma verified for *all fifteen* primes $R \equiv 3 \pmod 4$, $R \leq 107$ — by no residual among those fifteen is $O\big(x/(\log x)^{17/2}\big)$; (iv) a general density reduction: for every A there is a finite residual list whose joint exceptional set has relative density $O((\log x)^{-A})$ among hard primes. These densities are deliberately not the point: Vaughan's classical bound $\ll x \exp(-c(\log x)^{2/3})$ is asymptotically far stronger. What the chain contributes is *mechanism* — each bound is explicit, produces verifiable certificates, and rests on exact criteria and machine-checked lemmas rather than mean-value estimates. Computationally, we exhibit verified residual certificates for all 14,215,707 hard primes below 10^{10}: the maximal minimal residual is 107, attained at the single prime 8,803,369 and unchanged from 5×10^7 to 10^{10}; a fixed list of 18 residuals covers every hard prime below 10^9. The measured density of the joint $\{3, 7\}$-exceptional set is $\approx 5.17/\log x$, matching the sieve exponent exactly.

1 Introduction

Erdős and Straus conjectured that for every integer $n \geq 2$ the equation

$$\frac{4}{n} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \tag{1}$$

has a solution in positive integers a, b, c. By multiplicativity it suffices to treat prime n, and classical polynomial identities settle every prime outside the six residue classes

$$\mathcal{H} \bmod 840 = \{1,\ 121,\ 169,\ 289,\ 361,\ 529\},$$

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the squares of units modulo 840 (Mordell [4]). We call primes in these classes *hard primes.* Direct verification has confirmed the conjecture far beyond any range considered here: to 10^{14} by Swett [7] and to 10^{17} by Salez [5]; the present paper is logically independent of those bounds. The structure of the solution count was analyzed by Elsholtz and Tao [1]. Verification, however, exhibits no mechanism; the aim here is a quantitative structural reduction. For calibration, Vaughan's classical mean-value bound [6] gives an exceptional set $\ll x\exp(-c(\log x)^{2/3})$ for (1) — asymptotically stronger than any fixed power of $\log x$; the value of the present approach is not raw density but its explicit, certificate-generating, machine-verified structure, together with the exact criteria.

1.1 The residual method

Fix a hard prime p. All six classes satisfy

$$\text{(H1) } p \equiv 1 \pmod 8, \qquad \text{(H2) } p \equiv 1 \pmod 3, \qquad \text{(H3) } p \bmod 7 \in \{1,2,4\}. \tag{2}$$

For an *admissible residual* $R \equiv 3 \pmod 4$ put

$$a \;=\; \frac{p+R}{4} \in \mathbb{Z}, \qquad m \;=\; p\,a.$$

Substituting a into (1) and clearing denominators shows that a solution with first denominator a exists if and only if

$$\exists\, k \mid m^2 \quad \text{with} \quad k \equiv -m \pmod R, \tag{3}$$

in which case $b = (k+m)/R$ and $c = (m^2/k + m)/R$ are positive integers solving (1). We write $R_{\min}(p)$ for the least admissible R for which (3) holds. Two elementary congruences drive everything below. From $4a = p + R$,

$$a \equiv 4^{-1}p, \qquad m \equiv 4^{-1}p^2 \pmod R, \tag{4}$$

and since $\gcd(p,R) = 1$ (as $p > R$ is prime), also $\gcd(a,R) = \gcd(m,R) = 1$: the setting is always a unit computation modulo R. By (4), m is a *square* modulo every prime $r \mid R$; hence for every prime $r \mid R$ with $r \equiv 3 \pmod 4$ (at least one exists, since $R \equiv 3 \pmod 4$),

$$\left(\tfrac{-m}{r}\right) \;=\; \left(\tfrac{-1}{r}\right) \;=\; -1\,: \tag{5}$$

the target class $-m$ is a quadratic non-residue mod r.

1.2 Main results

Theorem 1.1 ($R = 3$; exact)**.** *Let $p \equiv 1 \pmod 4$ with $3 \nmid p$, and $a = (p+3)/4$. Then* (3) *holds for $R = 3$ if and only if a has a prime factor $q \equiv 2 \pmod 3$. When it does, $k = q$ is a certificate.*

Theorem 1.2 ($R = 7$, hard primes; exact)**.** *Let p be a hard prime and $a = (p+7)/4$. Then* (3) *holds for $R = 7$ if and only if a has a prime factor that is a quadratic non-residue modulo* 7*.*

Theorem 1.3 ($R = 11$, hard primes; exact)**.** *Let p be a hard prime and $a = (p+11)/4$; note $3 \mid a$, $2 \nmid a$, $7 \nmid a$ automatically. Then* (3) fails *for $R = 11$ if and only if either*

(a) every prime factor of a is a quadratic residue mod 11*; or*

(b) $v_3(a) = 1$, *every other prime factor of* a *is* $\equiv 1 \pmod{11}$ *except a non-residue part of exactly one of the shapes*

$$\{q \equiv 2\}, \qquad \{q \equiv 6\}, \qquad \{q \equiv 2\} \cup \{q' \equiv 6\} \pmod{11},$$

each with multiplicity one, paired respectively with $p \equiv 2,\ 6,\ 1 \pmod{11}$.

Theorems 1.2 and 1.3 are established by finite, machine-verified case analyses (Section 3); the reduction to a finite check is itself a theorem:

Remark 1.4 (computer-assisted components). The exhaustive verifications behind Theorems 1.2, 1.3 and Lemma 1.7 below are *parts of the respective proofs*, on the same footing as a hand case analysis. Each is a deterministic enumeration of a finite configuration space whose finiteness and sufficiency are proved in advance (Theorem 1.5; for Lemma 1.7, monotonicity of target-reachability in support and multiplicities, checked at minimal exponents). The checks are short (seconds on commodity hardware), involve only exact integer arithmetic, and the code is available in the accompanying repository; Appendix A records the exact configuration counts and entry points.

Theorem 1.5 (meta-theorem)**.** *For every fixed prime* $R \equiv 3 \pmod 4$*, the validity of* (3) *at a hard prime* p *depends only on* (i) *the multiset of classes mod* R *of the prime factors of* $a = (p+R)/4$*, with multiplicities capped at* $\lceil (R-1)/2 \rceil$*, and* (ii) $p \bmod R$*. Consequently an exact solvability criterion exists for every* R *and is computable by finite enumeration.*

The anchor result converts the exact criteria into an unconditional bound.

Theorem 1.6 (sieve bound)**.** *The number of hard primes* $p \le x$ *with* $R_{\min}(p) \ge 11$ *(that is, solved by neither* $R = 3$ *nor* $R = 7$*) is* $O\big(x/(\log x)^2\big)$.

The exponent grows by $\frac{1}{2}$ for each further residual with a sieve-grade handle. For $R = 11$ this handle is Theorem 1.3; for larger R it is the following machine-verified lemma (Section 4).

Lemma 1.7 (support bound; verified for every prime $R \equiv 3$ (4), $19 \le R \le 107$)**.** *Let* R *be one of the twelve primes* $19, 23, 31, 43, 47, 59, 67, 71, 79, 83, 103, 107$*. Every configuration for which residual* R *fails at a hard prime has at most* $(R-3)/2$ *nonzero factor-class support; equivalently, the prime factors of* $(p+R)/4$ *lie in at most* $(R-1)/2$ *of the* $R-1$ *unit classes mod* R*. Consequently failure at* R *forbids at least half the classes, through finitely many maximal-support branches.*

Theorem 1.8 (the chain)**.** *The number of hard primes* $p \le x$ *failing every residual in* $\{3, 7, 11\}$*, in* $\{3, 7, 11, 19\}$*, and in* $\{3, 7, 11, 19, 23\}$ *is, respectively,*

$$O\bigg(\frac{x}{(\log x)^{5/2}}\bigg), \qquad O\bigg(\frac{x}{(\log x)^{3}}\bigg), \qquad O\bigg(\frac{x}{(\log x)^{7/2}}\bigg).$$

More generally, with $\mathcal{P} = \{3, 7, 11, 19, 23, 31, 43, 47, 59, 67, 71, 79, 83, 103, 107\}$ *— all fifteen primes* $\equiv 3 \pmod 4$ *up to* 107 *— the number of hard primes failing every residual in* $\mathcal{P}$ *is*

$$O\bigg(\frac{x}{(\log x)^{17/2}}\bigg).$$

In particular all but a proportion $O((\log x)^{-2})$ *of hard primes are solved within* $\{3, 7, 11, 19\}$*, and all but* $O((\log x)^{-15/2})$ *within* $\mathcal{P}$.

Proposition 1.9 (aggregate identity families)**.** *Let $p \equiv 1 \pmod 4$ be prime and $R \equiv 3 \pmod 4$. If $R \mid p+1$, then $k = ap^2$ is a certificate for residual R; if $R \mid p+4$, then $k = a^2 p$ is. Hence any prime factor $\equiv 3 \pmod 4$ of $p+1$ or of $p+4$ yields an explicit solution, and these are the only a-independent divisor shapes $k = a^j p^i$ with this property. Adding the two corresponding linear forms to Theorem 1.8: the number of hard primes $p \le x$ failing every residual in $\mathcal{P}$ and both families is $O\big(x/(\log x)^{19/2}\big)$.*

Theorem 1.10 (density reduction)**.** *For every $A > 0$ there is a finite admissible set $S(A)$ such that the number of hard primes $p \le x$ for which every residual in $S(A)$ fails is $O_A\big(x/(\log x)^{1+A}\big)$.*

Remark 1.11 (relation to Vaughan's bound)*.* Theorem 1.10 and the chain of Theorem 1.8 are *weaker in density* than Vaughan's bound $\ll x \exp(-c(\log x)^{2/3})$ [6], which already exceeds every fixed power of $\log x$. Their interest is orthogonal: the exceptional sets here are cut out by finitely many *explicit* multiplicative conditions on shifted forms, each failure of which is constructively certified by an exhibited divisor k, and the underlying lemmas are exact or machine-verified. Vaughan's method counts solutions on average and exhibits neither certificates nor criteria.

Computations (Section 5) provide verified certificates for all hard primes below 10^{10} and calibrate every constant in sight.

2 The obstruction framework

Proposition 2.1 (character obstruction)**.** *Let $r \mid R$ be prime with $r \equiv 3 \pmod 4$. If every prime factor q of $m = pa$ satisfies $\big(\frac{q}{r}\big) = +1$, then (3) fails for R.*

Proof. Every divisor k of m^2 is a product of prime factors of m, so $\big(\frac{k}{r}\big) = +1$. A certificate requires $k \equiv -m \pmod r$, whence $\big(\frac{k}{r}\big) = \big(\frac{-m}{r}\big) = -1$ by (5) — impossible. ☐

We call failures produced by Proposition 2.1 *Type I*; failures where m possesses a non-residue prime factor modulo every relevant r, yet no divisor of m^2 attains the exact class $-m \bmod R$, are *Type II*.

Proposition 2.2 (guaranteed-success classes)**.** *Let $t \equiv -4^{-1}p^2 \pmod R$. If a has a prime factor q with $q \equiv t\,p^{-i} \pmod R$ for some $i \in \{0,1,2\}$, then $k = q\,p^i$ is a certificate. In particular, since $t\,p^{-2} \equiv -4^{-1} \pmod R$, any prime factor of a in the fixed class $-4^{-1} \bmod R$ certifies success regardless of p.*

Proof. $k = qp^i$ divides m^2 (as $q \mid a$ and $i \le 2$) and lies in class $t \equiv -m$. Integrality of b, c is automatic from $k \equiv -m$ and $m^2/k \equiv m^2(-m)^{-1} \equiv -m \pmod R$. ☐

The contrapositive of Proposition 2.2 is the sieve input for Theorem 1.10: failure at R forces $(p+R)/4$ to avoid prime factors in up to three explicit classes modulo R.

The next identity converts the per-residual character conditions into a single global statement about p, and is the structural explanation of joint failure across residuals.

Theorem 2.3 (reciprocity structure)**.** *Let p be a hard prime, $R \equiv 3 \pmod 4$ prime, and q an odd prime factor of $a = (p+R)/4$ (necessarily $q \ne R$). Then*

$$\Big(\frac{q}{R}\Big) \;=\; \Big(\frac{p}{q}\Big).$$

Proof. From $q \mid p+R$ we get $R \equiv -p \pmod{q}$. By quadratic reciprocity with $R \equiv 3 \pmod{4}$, $\left(\frac{q}{R}\right) = (-1)^{\frac{q-1}{2}} \left(\frac{R}{q}\right) = (-1)^{\frac{q-1}{2}} \left(\frac{-1}{q}\right) \left(\frac{p}{q}\right) = \left(\frac{p}{q}\right)$. □

Corollary 2.4. *(i) The hard classes are the squares mod* 840*, so* p *is automatically a quadratic residue mod* $3, 5, 7$*, and* $p \equiv 1 \pmod{8}$ *handles* $q = 2$ *(which divides* a *only for* $R \equiv 7 \pmod{8}$*, where* $\left(\frac{2}{R}\right) = +1$*):* the small primes forced into the shifted values by the Mordell structure never break a Type-I failure — *a conceptual explanation of why the hard classes are exactly the square classes. (ii)* p *fails Type-I at every prime residual* $R \le B$ *simultaneously if and only if* $\left(\frac{p}{R}\right) = +1$ *for each such* R *and* $\left(\frac{p}{q}\right) = +1$ *for every unforced odd prime* q *dividing any of the shifted values* — one condition per distinct prime, shared across every residual it divides. *This is the correlation mechanism behind the deep tail: smooth shifted values have fewer distinct prime factors, hence fewer independent sign conditions. For the record prime* 8,803,369*, the* 27 *shifted values contain only* 43 *distinct odd primes (typical:* 45–52*) and* 81% *of them satisfy* $\left(\frac{p}{q}\right) = +1$ *(typical:* 44–60%*) — a* $\approx 4\sigma$ *fluctuation in provably the right statistic. Verified:* 135,588 *instances of Theorem 2.3 and* 61,438 *forced-prime instances in the* 10^9 *data, zero violations of either.*

3 Exact criteria

3.1 Proof of Theorem 1.1

Modulo 3 the unit group has order 2, so quadratic character equals class. Since $p^2 \equiv 1 \pmod{3}$, (4) gives $m \equiv 1$ and the target is $-m \equiv 2 \pmod{3}$; also $3 \nmid a$. *Necessity:* if every prime factor of a is $\equiv 1 \pmod{3}$, then $a \equiv 1 \pmod{3}$, hence $p = 4a - 3 \equiv 1 \pmod{3}$ as well, so every prime factor of $m = pa$ is $\equiv 1 \pmod{3}$ and every divisor of m^2 is $\equiv 1 \pmod{3}$: class 2 is never attained. *Sufficiency:* $k = q \equiv 2 \pmod{3}$ divides m^2, and the recovery formulae give integers as in Proposition 2.2. □

3.2 Proof of Theorem 1.5

A certificate is a divisor $k \mid m^2$ in class $-m \bmod R$. The set of classes of divisors of m^2 is determined by the classes of the prime factors of $m = pa$ together with their multiplicities, each factor being usable with exponent at most twice its multiplicity; exponents are only meaningful modulo $\mathrm{ord}_R(\text{class}) \le R - 1$, so multiplicities beyond $\lceil (R-1)/2 \rceil$ are equivalent to that cap. The target class and the consistency relation $\prod(\text{classes}) \equiv a \equiv 4^{-1}p \pmod{R}$ are functions of $p \bmod R$ by (4). Hence solvability is a function of the capped class multiset and $p \bmod R$, a finite amount of data. □

3.3 Proof of Theorem 1.2

($\Rightarrow$) is Proposition 2.1 with $r = R = 7$. ($\Leftarrow$) reduces to a finite verification via Theorem 1.5. Write $(\mathbb{Z}/7)^* \cong \mathbb{Z}/6$ in discrete-log coordinates. Three structural constraints hold at hard primes:

(i) $p \bmod 7 \in \{1, 2, 4\}$, by (H3);

(ii) $2 \mid a$: by (H1), $p + 7 \equiv 0 \pmod{8}$, so $a = (p+7)/4$ is even — the class of 2 (a residue mod 7) is always present;

(iii) the factor classes multiply to $a \equiv 4^{-1}p \pmod 7$.

Enumerating all class multisets with multiplicities capped at 3 subject to (i)–(iii) yields 1536 consistent configurations; for each, one checks whether the target class is attained by a bounded-exponent subproduct. The verification (code: `theory.verify_R7_finite`) confirms: the target is attainable precisely when some factor class is a non-residue. □

Remark 3.1. The hard-class hypotheses are essential: for $p = 701 \equiv 1 \pmod 4$ (not hard), $a = 177 = 3 \cdot 59$ has the non-residue factor 3, yet $R = 7$ fails. Constraint (ii) — the forced even factor — is exactly what rescues every would-be Type II configuration at hard primes. Empirically (Section 5), among 41,421 sampled failures of $R = 7$ at hard primes, *all* are Type I.

3.4 Proof of Theorem 1.3

By Theorem 1.5 with $R = 11$, $d = 10$: multiplicities cap at 5, and at hard primes $3 \mid a$ (from (H2), $11 \equiv 2 \bmod 3$), while $2 \nmid a$ (from (H1), $p + 11 \equiv 4 \bmod 8$) and $7 \nmid a$ (from (H3), $-11 \equiv 3 \bmod 7 \notin \{1, 2, 4\}$). A dynamic-programming enumeration over the finite configuration space (code: `theory.finite_criterion_dp`) compresses it to 25 equivalence states: 16 succeed, 6 fail with all factor classes quadratic residues (case (a)), and exactly 3 fail with a non-residue present (case (b)); the three failing states decode to the three listed patterns. Two structural facts emerge from the enumeration. First, no state fails by a *proper-subgroup* obstruction: the only subgroup of $\mathbb{Z}/10$ containing an odd element properly is $\{0, 5\}$, and the consistency relation forces p's class to extend it to all of $\mathbb{Z}/10$. Second, the case-(b) failures are pure exponent-budget edge cases: any of $v_3(a) \geq 2$, a factor in class $\{4, 5, 9\}$, or a second non-residue factor restores solvability. The criterion agrees with ground-truth divisor search for 158,759 of 158,759 hard primes tested below 10^9. □

4 The sieve bound

Proof of Theorem 1.6. Step 1 (exact criteria). Let p be a hard prime and set $n = (p + 3)/4$, so that $(p + 7)/4 = n + 1$: *the two shifted forms are consecutive integers.* By Theorems 1.1 and 1.2, p is exceptional (both residuals fail) precisely when

$$n \text{ has no prime factor} \equiv 2 \pmod 3 \quad \text{and} \quad n + 1 \text{ has no prime factor} \equiv 3, 5, 6 \pmod 7,$$

with $p = 4n - 3$ prime.

Step 2 (congruence bookkeeping). Fix one of the six hard classes $h \bmod 840$; then $n \equiv (h + 3)/4 \pmod{210}$ is fixed, and the conditions at the primes $2, 3, 5, 7$ are determined by the class (in particular $3 \nmid n$ and $7 \nmid n + 1$ automatically). It suffices to bound each class and sum.

Step 3 (containment in a sifted set). Let $z = x^{1/10}$. After discarding the $O(z)$ primes $p \leq z$, every exceptional p gives $n \leq (x + 3)/4$, $n \equiv n_0 \pmod{210}$, such that for every prime q with $7 < q \leq z$:

(a) $q \equiv 2 \pmod 3 \implies n \not\equiv 0 \pmod q$;

(b) $q \equiv 3, 5, 6 \pmod 7 \implies n \not\equiv -1 \pmod q$;

(c) $q \nmid 4n - 3$ (since $4n - 3 = p$ is prime, $p > z$).

Conditions (a), (b) only use "no forbidden factor up to z", which is implied by the full criteria — an upper bound needs no more.

Step 4 (sieve of dimension 2). This is a standard upper-bound sieve for the three linear forms n, $n+1$, $4n-3$, with sifting function

$$\omega(q) \;=\; \mathbf{1}_{q\equiv 2(3)} \;+\; \mathbf{1}_{q\,\mathrm{NQR}\,(7)} \;+\; 1 \qquad (q>7),$$

whose forbidden classes $0,-1,3\cdot 4^{-1} \bmod q$ are pairwise distinct for $q\nmid 21$. By Dirichlet, ω has average $\kappa=\frac{1}{2}+\frac{1}{2}+1=2$ in the Halberstam–Richert sense $\sum_{q\le w}\omega(q)\log q/q=\kappa\log w+O(1)$. The Fundamental Lemma of sieve theory ([3, Thm. 2.5]; [2, Lem. 6.3]) with $z=x^{1/10}$ gives

$$\#E(x) \;\ll\; x\prod_{7<q\le z}\Bigl(1-\frac{\omega(q)}{q}\Bigr) \;\asymp\; x\,(\log z)^{-1}(\log z)^{-1/2}(\log z)^{-1/2} \;\asymp\; \frac{x}{(\log x)^2},$$

by Mertens' theorem and its arithmetic-progression form. Summing over the six classes preserves the bound.

□

The verification behind Lemma 1.7 reduces to finitely many checks through the following elementary but load-bearing observation, which we isolate for scrutiny.

Lemma 4.1 (monotonicity reduction)**.** *For a class multiset C (classes v with multiplicities μ_v) let $D(C)=\{\sum_v e_v v : 0\le e_v\le 2\mu_v\}$ denote the set of attainable divisor logs (before the p-shifts). If $\mu_v\le\mu'_v$ for all v, then $D(C)\subseteq D(C')$; likewise enlarging the support enlarges D. Consequently a support S carries* some *failing configuration (any multiplicities ≥ 1 on S) if and only if the all-multiplicity-one configuration on S fails.*

Proof. Any admissible exponent vector for C is admissible for C', giving the inclusions. If a configuration on S with multiplicities $\mu_v\ge 1$ fails, the target lies outside $D(C)$, hence outside the subset $D(C_1)$ attained at multiplicities one — so C_1 fails too; the converse inclusion is trivial. The same argument applies verbatim after adjoining the p-shifts. □

Proof of Lemma 1.7. By Lemma 4.1 it suffices to check that for every support of size $(R-1)/2$ over the nonzero discrete logs and every class of p mod R, the target is reachable already at minimal multiplicities (exponent budget 2 per class, p-budget 2); success there kills every configuration on every superset support at any multiplicities. The check deliberately ignores the consistency relation $\prod(\text{classes})\equiv 4^{-1}p$: it therefore examines a *superset* of the realizable configurations, and can only overstate the failing supports — the lemma is conservative. For $R=19,23$ a direct enumeration of all $\binom{R-2}{(R-1)/2}$ supports suffices (437,580 resp. 7,759,752 pairs; code: `theory.verify_support_bound`). For the larger residuals the binomial count explodes, and the check is instead performed by a subset dynamic program (`theory.verify_support_bound_dp`): fold all supports into a table mapping each reachability mask to the maximal support size achieving it — valid because, by the same monotonicity, failure at any multiplicities implies failure at multiplicity one. The realized masks are heavily structured, so the state count stays tractable (3,001 states at $R=31$ against 77.5 million supports; 19,530,971 states at $R=107$, 331 seconds). All twelve residuals verify with zero violations; exact state counts are tabulated in Appendix A. The bound is tight: the all-residue (Type I) configurations occupy exactly $(R-3)/2$ nonzero classes. □

Proof of Theorem 1.8. Write $n = (p+3)/4$; the relevant forms are n, $n+1$, $n+2 = (p+11)/4$, $n+4 = (p+19)/4$, $n+5 = (p+23)/4$, and $4n-3 = p$. Joint failure of $\{3, 7, 11\}$ implies, by Theorems 1.1, 1.2, 1.3: n free of primes $\equiv 2 \pmod 3$; $n+1$ free of non-residues mod 7; and, splitting by the two cases of Theorem 1.3: *branch (a)* — $n+2$ free of the five non-residue classes mod 11 (density $\frac{1}{2}$); *branch (b)* — $n+2$ free of the six classes $\{4, 5, 7, 8, 9, 10\}$ mod 11 (density $\frac{3}{5}$). Each branch is a four-form sieve problem of dimension $1+\frac{1}{2}+\frac{1}{2}+\frac{1}{2} = \frac{5}{2}$, resp. $1+\frac{1}{2}+\frac{1}{2}+\frac{3}{5} = \frac{13}{5}$. Since $\frac{13}{5} > \frac{5}{2}$, the Type-II branch contributes $O\big(x/(\log x)^{13/5}\big) = o\big(x/(\log x)^{5/2}\big)$: the bound is the maximum of the branch exponents, attained by the character branch (a). The Fundamental Lemma applied to each branch gives $O(x/(\log x)^{5/2})$ in total.

For $\{3, 7, 11, 19\}$, Lemma 1.7 splits failure at 19 into finitely many maximal-support branches, each forbidding an explicit set of at least 9 of the 18 unit classes mod 19 on $n+4$ — dimension $\geq \frac{1}{2}$. Crossing with the branches at 11 leaves finitely many combined branches, each a five-form problem of dimension ≥ 3 (those involving the Type-II case at 11 have dimension $\geq 1+\frac{1}{2}+\frac{1}{2}+\frac{3}{5}+\frac{1}{2} = \frac{31}{10} > 3$ and are again lower-order); the forbidden residues of the five forms are pairwise distinct for $q > 19$, smaller primes being absorbed into the congruence class of n. Summing the finitely many branches gives $O(x/(\log x)^3)$. The $\{3, 7, 11, 19, 23\}$ case adds the form $n+5$ with Lemma 1.7 at $R = 23$, one more factor of $(\log x)^{1/2}$. The full chain $\mathcal{P}$ is identical: the fifteen forms $n + (R-3)/4$, $R \in \mathcal{P}$, are distinct integer shifts of n (pairwise differences less than 27), each contributing dimension $\geq \frac{1}{2}$ through finitely many maximal-support branches, plus 1 for primality: total dimension $\geq 1 + \frac{15}{2} = \frac{17}{2}$, with forbidden residues pairwise distinct for $q > 107$ and smaller primes absorbed into the congruence class of n. □

Proof of Proposition 1.9. $k = ap^2$ divides $m^2 = p^2a^2$ and $k \equiv -m$ iff $ap^2 \equiv -pa$, i.e. $p \equiv -1 \pmod R$; $k = a^2p$ divides m^2 and $a^2p \equiv -pa$ iff $a \equiv -1$, i.e. $p \equiv -4 \pmod R$ by (4). Integrality of b, c is automatic. Scanning all shapes $k = a^jp^i$ $(j, i \leq 2)$ yields no further p-only condition. The failure of both families states that the two forms $(p+1)/2$ and $p+4$ (in n: $2n-1$ and $4n+1$) are free of primes $\equiv 3 \pmod 4$ — two further sifting conditions of dimension $\frac{1}{2}$ each on forms distinct from all previous ones. Empirically the two families alone cover 74% of hard primes below 10^9, and none of the four critical primes. □

Remark 4.2. Each further prime residual $R \equiv 3 \pmod 4$ whose support-bound lemma is verified adds $\frac{1}{2}$ to the exponent; the dynamic-program verification has now been carried through every prime residual up to 107. The chain is a concrete instantiation of Theorem 1.10 with explicit, verified constants. Two caveats frame its scope. First, the composite residuals $(15, 27, 35, \ldots)$ are not covered by the present lemma; treating them requires the componentwise theory mod each prime-power factor. Second — and fundamentally — no fixed finite list can be pushed to *emptiness* by upper-bound sieving: the count bound $x/(\log x)^\kappa$ diverges for every fixed κ, and the calibrated model of Section 5 predicts a positive density constant at every fixed list, i.e. infinitely many exceptional primes with the record minimal residual growing slowly without bound. The correctly-posed remaining problem is a slowly growing bound $R_{\min}(p) \ll f(p)$, which requires existence (lower-bound) technology beyond upper-bound sieves.

Proof of Theorem 1.10. Fix $A > 0$. Choose $B = B(A)$ so large that

$$\sum_{\substack{3 \leq R \leq B \\ R \equiv 3\,(4)}} \frac{1}{\varphi(R)} \;\geq\; \sum_{\substack{3 \leq R \leq B \\ R \equiv 3\,(4)}} \frac{1}{R} \;>\; A; \tag{6}$$

this is possible since the reciprocal sum over the progression $R \equiv 3 \pmod 4$ diverges, and $\varphi(R) \le R$. Let $S(A) = \{R \equiv 3\,(4) : 3 \le R \le B\}$ and $M = 840 \cdot \prod_{R\in S(A)} R$.

Write $n = (p+3)/4$ as before, so $(p+R)/4 = n + (R-3)/4$ for each $R \in S(A)$: a family of $|S(A)|$ integer linear forms in n with pairwise distinct shifts, together with the form $4n-3 = p$. Fix a residue class $n \equiv n_0 \pmod M$ compatible with p hard; there are finitely many, and it suffices to bound each. Within such a class, for every $R \in S(A)$ the quantities $t \equiv -4^{-1}p^2$ and p^{-1} mod R are constant, so the class set $S_R(p) = \{t, tp^{-1}, tp^{-2}\}$ mod R of Proposition 2.2 is a fixed nonempty set of $t_R = t_R(n_0) \in \{1,2,3\}$ classes. By the contrapositive of Proposition 2.2, if every residual in $S(A)$ fails at p, then for each $R \in S(A)$ the form $n + (R-3)/4$ has *no* prime factor in $S_R(p)$; and $4n-3$, being prime and $> z$, has no prime factor $\le z$.

Sift the class $\{n \le (x+3)/4 : n \equiv n_0\ (M)\}$ by the primes $q \le z = x^{1/10}$, $q \nmid M$. A prime q forbids: the class $n \equiv 3\cdot 4^{-1} \pmod q$ (from the primality of $4n-3$), and, for each $R \in S(A)$ with q mod $R \in S_R(p)$, the single class $n \equiv -\frac{R-3}{4} \pmod q$ (a prime factor q of the form $n+(R-3)/4$ in a guaranteed-success class cannot occur). Thus

$$\omega(q) \;=\; 1 \;+\; \#\big\{R \in S(A) : q \bmod R \in S_R(p)\big\},$$

which is bounded by $|S(A)| + 1$; the forbidden residues of distinct forms are pairwise distinct for $q > B$, since the shifts $(R-3)/4$ differ by less than B and $q \nmid M$. For each R, the density of primes q with q mod $R \in S_R(p)$ is $t_R/\varphi(R)$ by Dirichlet's theorem, so ω has average

$$\kappa \;=\; 1 + \sum_{R\in S(A)} \frac{t_R}{\varphi(R)} \;\ge\; 1 + \sum_{R\in S(A)} \frac{1}{\varphi(R)} \;>\; 1 + A$$

in the Halberstam–Richert sense, by (6). The Fundamental Lemma ([3, Thm. 2.5]) in dimension κ then bounds the sifted count by $O_A\Big(x\prod_{q\le z}(1-\omega(q)/q)\Big) \;=\; O_A(x(\log x)^{-\kappa}) \;=\; O_A\big(x/(\log x)^{1+A}\big)$, and summing over the finitely many classes n_0 mod M preserves the bound. □

Remark 4.3. Theorem 1.10 is the strongest statement the pure sieve can give: density $O((\log x)^{-A})$ for every A, but not emptiness. The conjecture for hard primes is equivalent to a *finite covering hypothesis*: some finite S admits, for every hard prime, a residual $R \in S$ with (3). All computational evidence below supports $S = \{3, 7, 11, \ldots, 107\}$.

5 Computations to 10^{10}

All code and data are in the accompanying repository. Certificates are found by direct integer factorization of a (trial division; the factorization of $m^2 = p^2a^2$ follows) and scanning divisors for the class condition (3); every certificate is re-verified in exact integer arithmetic, $4abc = n(bc + ac + ab)$.

5.1 Data

Every one of the 14,215,707 hard primes $p < 10^{10}$ possesses a residual certificate with $R \le 107$. The minimal residuals distribute as follows (10^{10} data):

R	count	R	count	R	count
3	7 386 405	31	30 222	63	57
7	3 472 188	35	4 775	67	18
11	2 325 588	39	4 307	71	46
15	487 889	43	1 131	75	3
19	322 871	47	1 661	79	11
23	150 828	51	240	83	5
27	27 000	55	263	107	1
		59	198		

Salient features:

- $R_{\max} = 107$, attained by the single prime 8,803,369 $< 10^7$, *unchanged from* 5×10^7 *to* 10^{10} — three orders of magnitude.
- No prime has minimal $R \in \{87, 91, 95, 99, 103\}$. This gap requires no modular explanation: under the calibrated independence model the expected counts there are 0.12, 0.03, 0.05, 0.01, 0.01. Conditioned on failing every $R \leq 83$, a prime lands at 87 with probability 0.53 and at 107 with probability 0.017; the record prime is the anomaly, not the gap.
- Among all 1,587,581 hard primes below 10^9, exactly *four* admit a unique working residual ≤ 107 (8,803,369 with 107; 142,361,209 with 59; 287,567,281 with 83; 794,037,841 with 63) — the entire obstruction to a shorter covering list. A greedy cover of the full solvability table uses 18 residuals; a packing argument gives a lower bound of 12.

5.2 Calibration of Theorem 1.6

The proportion of hard primes failing both $R = 3$ and $R = 7$, multiplied by $\log x$, is constant to within 2% across three decades of the 10^9 data:

bin ($p \sim$)	10^6	10^7	6×10^7	2.5×10^8	10^9
density $\times \log x$	5.14	5.19	5.18	5.17	5.16

That is, the joint exceptional density is $\approx 5.17/\log x$: the sieve exponent of Theorem 1.6 is exactly right, and 5.17 estimates the (conjectural) Selberg–Delange constant. The exceptional set is density zero but decays slowly: 26.5% of hard primes below 10^9 fail both residuals; all are settled by deeper residuals of the list. The fitted per-residual decay exponents likewise match the sieve prediction $\kappa = \frac{1}{2}$ where exact criteria exist (e.g. $\hat{\kappa} = 0.539$ for $R = 7$).

5.3 Failure taxonomy

Sampling the complete solvability table at 10^9: failures at $R = 7$ are 100% Type I (as Theorem 1.2 requires); at $R = 11$, 89.7% Type I and 10.3% the budget pattern of Theorem 1.3(b); the Type I share decreases with $\varphi(R)$ (e.g. 51% at $R = 103$), as hitting one specific class among $\varphi(R)$ grows harder while the character dichotomy stays binary.

6 Open problems

1. **Finite covering.** Prove that some finite admissible set S covers all hard primes (Theorem 1.10 gives density $(\log x)^{-A}$ for every A; emptiness is beyond the sieve).

2. **Beyond the prime chain.** The support-bound lemma is now verified for every prime residual up to 107 (exponent $\frac{17}{2}$); extending past 107 meets the growth of the DP state space ($\approx 2\times$ per residual), and the composite residuals require the componentwise theory. More importantly, the fixed-list format itself cannot reach emptiness (see the remark after Theorem 1.8): the structurally right target is a slowly growing bound on $R_{\min}(p)$.

3. **The record and the covering hypothesis.** Theorem 2.3 explains the record prime quantitatively (see Corollary 2.4) and yields a conditional resolution of the covering question: under Dickson's prime-tuple conjecture, for every B there are infinitely many hard primes failing every residual $\leq B$ — fix the forced smooth parts of the shifted values (character-neutral by Corollary 2.4(i)), make the cofactors prime in prescribed residue classes, and apply Proposition 2.1. So no fixed finite list suffices, conditionally, and the record grows without bound (at the pure-Type-I level, like $\log x / \log\log x$). Unconditional versions appear out of reach: already "infinitely many hard p fail $R = 3$" contains a two-form prime-tuple statement. What remains open is an unconditional upper bound on $R_{\min}(p)$ — and this is not an incremental item: the residual formulation is *complete* (any solution with denominators $a \leq b \leq c$ has $p/4 < a \leq 3p/4$, so $R = 4a - p \in (0, 2p]$ with $R \equiv 3 \pmod 4$ automatic), hence $R_{\min}(p) \leq 2p$ is *equivalent* to the conjecture at p. An unconditional bound $R_{\min}(p) \leq f(p)$ for any f whatsoever is the full problem.

A Verification details

This appendix documents every computer-assisted claim of the paper: the finite enumerations that are *proof components* (Appendix A.1), the large-scale certificate computations behind Section 5 and how each dataset is verified (Appendix A.2), and the commands that reproduce all of it (Appendix A.3). All code is in the accompanying repository; every check is deterministic and uses exact integer arithmetic only.

A.1 Proof components

Each component below is an exhaustive enumeration of a finite space whose sufficiency is established in the text (Theorem 1.5 for the exact criteria; for Lemma 1.7, monotonicity of the reachability of the target in support and multiplicities, checked at minimal exponents). Each run completes in seconds on commodity hardware; the entry points named are in the module `erdos_straus.theory`.

component	configuration space	count	entry point
Thm. 1.2 ($R = 7$)	class multisets in $(\mathbb{Z}/7)^*$, multiplicities ≤ 3, with the forced even factor, $p \bmod 7 \in \{1, 2, 4\}$, and the consistency relation	1,536	`verify_R7_finite`
Thm. 1.3 ($R = 11$)	dynamic program over $(\mathbb{Z}/11)^*$ with the forced factor 3; collapses to equivalence states (16 succeed, 6 fail by character, 3 by budget)	25 states	`finite_criterion_dp`
Lem. 1.7 ($R = 19$)	all supports of size 9 over the nonzero logs $\times$ all classes of p, at minimal multiplicities	437,580	`verify_support_bound`
Lem. 1.7 ($R = 23$)	all supports of size 11, likewise	7,759,752	`verify_support_bound`
Lem. 1.7 (R = 19–107, all twelve primes)	subset dynamic program over reachability-mask states; state counts 271 (R=19), 3,001 (R=31), 2,414,145 (R=83), 19,530,971 (R=107); zero violations throughout	$\leq$ 19.5M states	`verify_support_bound_dp`

In addition (as sanity checks, not proof components), the criteria of Theorems 1.1–1.3 were compared against ground-truth divisor search on the complete solvability data below 10^9: agreement in 1,587,581/1,587,581 cases for $R = 3$ and 158,759/158,759 sampled cases for $R = 11$, with zero discrepancies.

A.2 Certificate datasets and their verification

The computational claims of Section 5 rest on the following datasets, all archived in the repository under `data/`. A certificate (R, a, b, c) for p is *accepted* only if $4abc = p(bc + ac + ab)$ holds in exact integer arithmetic and $R = 4a - p$; for the compact minimal-R maps, the verifier *reconstructs* the triple from (p, R) by the divisor search of (3) and then applies the same exact check, so a map entry is verified as a genuine certificate, not taken on faith.

dataset	entries	verification performed
explicit triples, $p < 1.2 \times 10^8$	213,131	every triple checked exactly
minimal-R map, $p < 10^9$	1,587,581	*exhaustive*: every entry reconstructed and checked exactly
minimal-R map, $p < 10^{10}$	14,215,707	143,000 systematically sampled entries reconstructed and checked; additionally *all* 842 tail entries ($R \geq 51$) checked with full minimality (no smaller admissible R succeeds)
solvability masks, $p < 10^9$	$27 \times 1{,}587{,}581$	per-prime solvability of every admissible $R \leq 107$ by full divisor-class computation; consistency with the minimal-R map

The masks table is the basis for the covering-set statements (the 18-residual cover, the lower bound 12, the four critical primes), the failure taxonomy, and the calibration of Section 5; the minimal-R maps are the basis for the distribution table, the record, and the gap analysis.

A.3 Reproduction

With the repository installed (`pip install -e .`), the following commands regenerate and re-verify everything above. Generation of the full 10^{10} map takes about 11 minutes on four cores; every other command runs in seconds to minutes.

claim	command
Thm. 1.2 finite check	python -c "from erdos_straus.theory import verify_R7_finite; print(verify_R7_finite())"
Thm. 1.3 enumeration	finite_criterion_dp(11, ...) as in the test suite (tests/test_solver.py)
Lem. 1.7 at $R = 19, 23$	verify_support_bound(19), (23)
generate the 10^{10} map	python -m erdos_straus.bulk_generate --max 10000000000 --segmented --store rmap
verify a dataset exhaustively	python -m erdos_straus.verify data/<file>
distribution / covering / taxonomy	python -m erdos_straus.analyze {dist, cover, tail}; python -m erdos_straus.theory

The full test suite (`python -m pytest`; 43 tests) exercises the proof-component checks, the exact criteria against ground truth, and sampled dataset validation on every run.

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